Mobile Game "Gate" Placement A/B Testing

Introduction

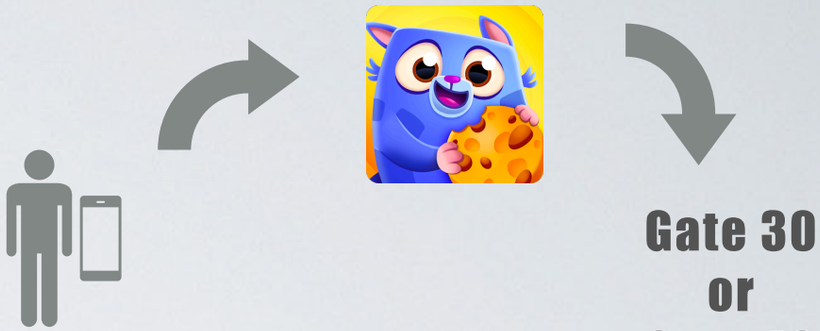
Mobile games frequently incorporate "gates" into their games. As a player progresses past certain levels, they will be forced to either wait a certain period of time or pay in order to continue through the gate. This gate serves two primary purposes:

1. Increase in-app purchases
2. Increase player satisfaction through forced breaks

The data for this analysis comes from DataCamp: "Mobile Games A/B Testing with Cookie Cats".

Data Collection

As players download the mobile application, they are randomly assigned to a different first gate level. Their data for the first week of playing was then collected. See graphic to the right:



Data Collected

- user id
- gate version
- total games played
- 1 day retention (T/F)
- 7 day retention (T/F)

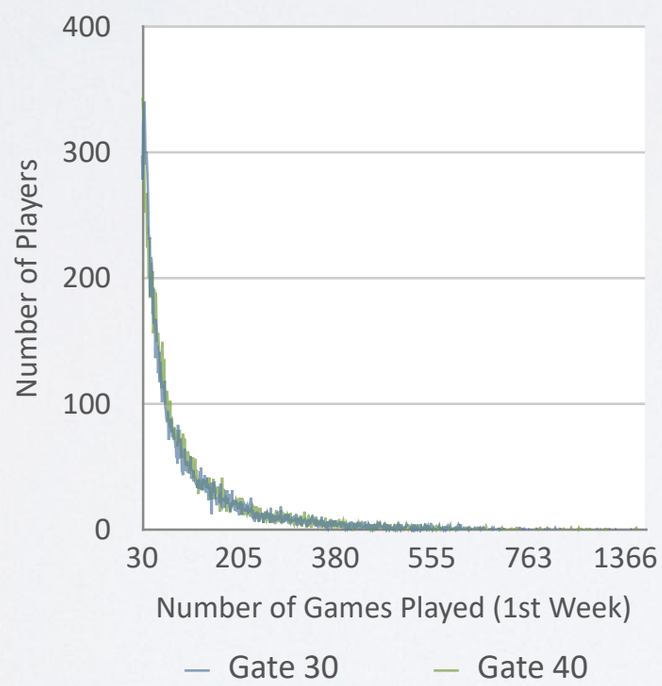
Analysis Goals

The focus of this A/B test is to use weekly games played and retention rates to determine which level this gate should be placed (level 30 or level 40) for the popular mobile game Cookie Cats.

Player Counts per Gate

- Players were evenly distributed between the two levels.
- Due to the number of players who never reached the minimum level for a gate, they will need to be excluded.

Player Counts per Level



- There does not appear to be a significant difference between the different gates and how many levels people play in their first week. Statistical analysis will provide greater insight.

Retention by Gate and Day

1 Day	Count	Total	Retention Rate	***7 Day***	Count	Total	Retention Rate
Gate 30	13,341	16,655	44.48%	**Gate 30**	6,145	16,655	26.95%
Gate 40	13,321	16,613	44.5%	**Gate 40**	6,028	16,613	26.62%
2 Sample Z Test for Proportions p-value			**0.85**	2 Sample Z Test for Proportions p-value			**0.11**

Conclusion

Retention rates seem to be unaffected by which level the gate is placed at. While the 7 day retention appears to favor the gate at level 30, the statistical significance is still relatively low with a p-value of 0.11. Additional gate options may be useful in determining the most effective level to place the gate.